GEMSTAR / ARM-FPGA RESEARCH

TASK 16 - FINAL RESEARCH REPORT

Literature Review, Research-Gap Analysis, Novelty Risk and Experimental Methodology

PRELIMINARY LITERATURE ASSESSMENT

Research artifact	Count
Literature publications	20
Comparison matrix entries	20
Priority competing studies	5
Novelty-risk areas	7
Experimental baselines	4
Research hypotheses	4
Evaluation metrics	7
Processing operations	5
Workflow stages	8

1. Research objective

Investigate whether sensor-derived reliability and tracking uncertainty can improve energy-aware ARM-FPGA processing decisions for Camera, LiDAR and Radar perception while satisfying timing and quality requirements.

This report documents the research design, not experimental proof of novelty or hardware performance.

2. Literature distribution

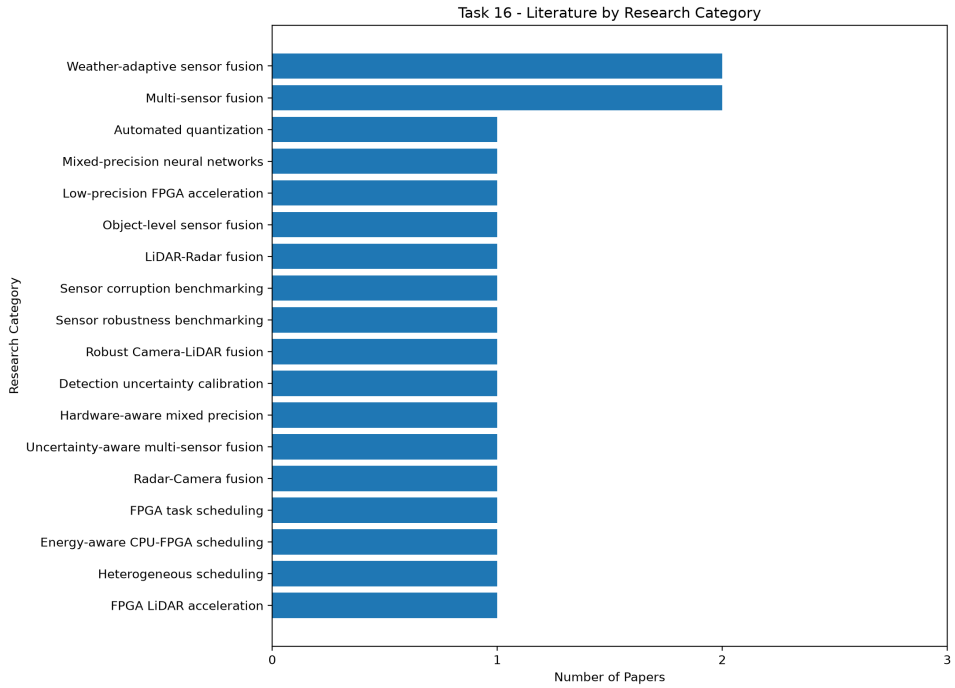


Figure 1. Distribution of the 20 collected papers by research category.

3. Literature inventory

P001 | Hardware-Accelerated 3D LiDAR-Based Object Detection with BEV Spatial Mapping on Embedded FPGA Platforms (2026)

Category: FPGA LiDAR acceleration | Venue: Electronics

Source: https://doi.org/10.3390/electronics15112296

Review: Metadata and abstract checked; full review pending

P002 | Dynamic contention-aware workflow scheduling on shared bus-based CPU-FPGA heterogeneous computing systems (2026)

Category: Heterogeneous scheduling | Venue: Expert Systems with Applications

Source: https://doi.org/10.1016/j.eswa.2025.130421

Review: Metadata and abstract checked; full review pending

P003 | BEVFusion: Multi-Task Multi-Sensor Fusion with Unified Bird's-Eye View Representation (2022)

Category: Multi-sensor fusion | Venue: ICRA

Source: https://arxiv.org/abs/2205.13542

Review: Abstract checked; full review pending

P004 | Multi-Modal Fusion Transformer for End-to-End Autonomous Driving (2021)

Category: Multi-sensor fusion | Venue: CVPR

Source: https://arxiv.org/abs/2104.09224

Review: Abstract checked; full review pending

P005 | Energy-aware fully-adaptive resource provisioning in collaborative CPU-FPGA cloud environments (2023)

Category: Energy-aware CPU-FPGA scheduling | Venue: Journal of Parallel and Distributed Computing

Source: https://doi.org/10.1016/j.jpdc.2023.02.009

Review: Abstract checked; full review pending

P006 | Energy-Aware Synchronization of Hardware Tasks in Virtualized Embedded Systems (2024)

Category: FPGA task scheduling | Venue: FPL 2024

Source: https://doi.org/10.1109/FPL64840.2024.00046

Review: Bibliographic information checked; full review pending

P007 | Radar-Camera Fusion in Perspective View and Bird's Eye View for 3D Object Detection (2025)

Category: Radar-Camera fusion | Venue: Sensors

Source: https://doi.org/10.3390/s25196106

Review: Publisher results checked; author metadata and full review pending

P008 | Seeing Through Fog Without Seeing Fog: Deep Multimodal Sensor Fusion in Unseen Adverse Weather (2020)

Category: Uncertainty-aware multi-sensor fusion | Venue: CVPR 2020

Source: https://openaccess.thecvf.com/content_CVPR_2020/html/Bijelic_Seeing_Through_Fog_Without_Seeing_Fog_Deep_Multimodal_Sensor_Fusion_CVPR_2020_paper.html

Review: Publisher abstract verified; full review pending

P009 | Towards Robust 3D Object Detection with LiDAR and 4D Radar Fusion in Various Weather Conditions (2024)

Category: Weather-adaptive sensor fusion | Venue: CVPR 2024

Source: https://doi.org/10.1109/CVPR52733.2024.01436

Review: Publisher abstract verified; full review pending

P010 | L4DR: LiDAR-4DRadar Fusion for Weather-Robust 3D Object Detection (2025)

Category: Weather-adaptive sensor fusion | Venue: AAAI 2025

Source: https://doi.org/10.1609/aaai.v39i4.32397

Review: Publisher abstract verified; full review pending

P011 | HAQ: Hardware-Aware Automated Quantization With Mixed Precision (2019)

Category: Hardware-aware mixed precision | Venue: CVPR 2019

Source: https://doi.org/10.1109/CVPR.2019.00881

Review: Publisher abstract verified; full review pending

P012 | Uncertainty Evaluation of Object Detection Algorithms for Autonomous Vehicles (2021)

Category: Detection uncertainty calibration | Venue: Automotive Innovation

Source: https://doi.org/10.1007/s42154-021-00154-0

Review: Publisher article checked; detailed review pending

P013 | TransFusion: Robust LiDAR-Camera Fusion for 3D Object Detection with Transformers (2022)

Category: Robust Camera-LiDAR fusion | Venue: CVPR 2022

Source: https://arxiv.org/abs/2203.11496

Review: Source metadata and abstract checked; full review pending

P014 | Benchmarking the Robustness of LiDAR-Camera Fusion for 3D Object Detection (2023)

Category: Sensor robustness benchmarking | Venue: CVPR Workshops 2023

Source: https://openaccess.thecvf.com/content/CVPR2023W/E2EAD/html/Yu_Benchmarking_the_Robustness_of_LiDAR-Camera_Fusion_for_3D_Object_Detection_CVPRW_2023_paper.html

Review: Source metadata and abstract checked; full review pending

P015 | Benchmarking Robustness of 3D Object Detection to Common Corruptions (2023)

Category: Sensor corruption benchmarking | Venue: CVPR 2023

Source: https://openaccess.thecvf.com/content/CVPR2023/html/Dong_Benchmarking_Robustness_of_3D_Object_Detection_to_Common_Corruptions_CVPR_2023_paper.html

Review: Source metadata and abstract checked; full review pending

P016 | Bi-LRFusion: Bi-Directional LiDAR-Radar Fusion for 3D Dynamic Object Detection (2023)

Category: LiDAR-Radar fusion | Venue: CVPR 2023

Source: https://openaccess.thecvf.com/content/CVPR2023/html/Wang_Bi-LRFusion_Bi-Directional_LiDAR-Radar_Fusion_for_3D_Dynamic_Object_Detection_CVPR_2023_paper.html

Review: Source metadata and abstract checked; full review pending

P017 | ObjectFusion: Multi-modal 3D Object Detection with Object-Centric Fusion (2023)

Category: Object-level sensor fusion | Venue: ICCV 2023

Source: https://openaccess.thecvf.com/content/ICCV2023/html/Cai_ObjectFusion_Multi-modal_3D_Object_Detection_with_Object-Centric_Fusion_ICCV_2023_paper.html

Review: Source metadata and abstract checked; full review pending

P018 | FINN: A Framework for Fast, Scalable Binarized Neural Network Inference (2017)

Category: Low-precision FPGA acceleration | Venue: FPGA 2017

Source: https://arxiv.org/abs/1612.07119

Review: Source metadata and abstract checked; full review pending

P019 | PACT: Parameterized Clipping Activation for Quantized Neural Networks (2018)

Category: Mixed-precision neural networks | Venue: arXiv preprint

Source: https://arxiv.org/abs/1805.06085

Review: Source metadata and abstract checked; full review pending

P020 | ReLeQ: A Reinforcement Learning Approach for Deep Quantization of Neural Networks (2018)

Category: Automated quantization | Venue: arXiv preprint

Source: https://arxiv.org/abs/1811.01704

Review: Source metadata and abstract checked; full review pending

4. Literature comparison matrix

The following is a compact summary of the complete research comparison CSV. 'Not established' signifies insufficient reviewed evidence, not confirmed absence.

ID	Category	FPGA	Uncertainty-based placement
P001	FPGA LiDAR acceleration	YES	Not established
P002	CPU-FPGA scheduling	YES	Not established
P003	Multi-sensor fusion	Not established	Not established
P004	Multi-sensor fusion	Not established	Not established
P005	Energy-aware CPU-FPGA scheduling	YES	Not established
P006	FPGA task scheduling	YES	Not established
P007	Radar-Camera fusion	Not established	Not established
P008	Uncertainty-aware multi-sensor fusion	Not established	Not established
P009	Weather-adaptive sensor fusion	Not established	Not established
P010	Weather-adaptive sensor fusion	Not established	Not established
P011	Hardware-aware mixed precision	Not established	Not established
P012	Detection uncertainty calibration	Not the stated focus	Not established
P013	Robust Camera-LiDAR fusion	Not established	Not established
P014	Sensor robustness benchmarking	Not established	Not established
P015	Sensor corruption benchmarking	Not established	Not established
P016	LiDAR-Radar fusion	Not established	Not established
P017	Object-level sensor fusion	Not established	Not established
P018	Low-precision FPGA acceleration	YES	Not established
P019	Mixed-precision neural networks	Not established	Not established
P020	Automated quantization	Not established	Not established

5. Priority competing research

ID	Study / topic	Priority
CR001	Context-aware FPGA sensor fusion	CRITICAL
CR002	Deadline-aware ADAS scheduling	HIGH
CR003	Age-of-information-aware scheduling	HIGH
CR004	Heterogeneous accelerator scheduling	HIGH
CR005	Uncertainty-aware multiprocessor scheduling	CRITICAL

6. CARMA full-text findings

Feature	Finding	Pages
Tri-sensor perception	Camera LiDAR and Radar used	3-5
ARM CPU and FPGA	ARM CPU hosts control and fusion; FPGA DPU performs inference	3-4
Dynamic reconfiguration	Context-based model branch and sensor activation	3-4
Energy optimization	Joint optimization of energy and object detection performance	2-5
Tracking uncertainty constraint	Tracking covariance constraint not demonstrated in reviewed formulation	2-5
Dynamic ARM-FPGA operator placement	Operator-level CPU versus FPGA assignment not demonstrated	3-5
Hardware measurement	Kria KV260 used for latency and energy experiments	4-5

7. CARMA versus DyUnS versus proposed system

Feature	CARMA	DyUnS	Proposed
Research objective	Context-aware energy-efficient sensor fusion	Uncertainty-aware multiprocessor scheduling	Uncertainty-constrained ARM-FPGA perception scheduling
Uncertainty source	Driving context	Uncertainty in computational model parameters	Sensor reliability and tracking uncertainty
Hardware	FPGA DPU system	Heterogeneous ARM multicore model	ARM CPU plus FPGA accelerators
Adaptation mechanism	Sensor and model-path activation	Uncertainty-aware task-to-core scheduling	Selected operator placement on ARM or FPGA
Optimization criteria	Energy and perception performance	Time power temperature reliability	Energy latency tracking quality deadlines
Tracking uncertainty constraint	Not established in current review	Not established in current review	Planned
Hardware evidence	FPGA prototype evaluation	Scheduling simulation	Future hardware measurements required

The comparison represents our present reading of the published work. Claims about missing capabilities remain provisional until all relevant methods are reviewed.

8. Novelty-risk assessment

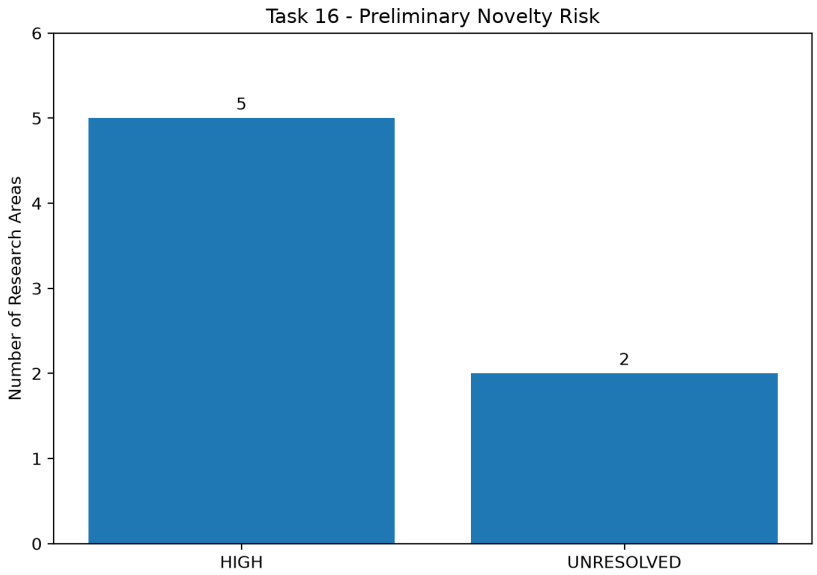


Figure 2. Preliminary classification of seven candidate research areas.

ID	Research area	Risk
N01	Tri-sensor fusion	HIGH
N02	Adaptive FPGA perception	HIGH
N03	CPU-FPGA scheduling	HIGH
N04	Uncertainty-aware task scheduling	HIGH
N05	Sensor-derived uncertainty for ARM-FPGA placement	UNRESOLVED
N06	Tracking-quality and deadline constraints	UNRESOLVED
N07	Adaptive mixed precision	HIGH

9. Candidate research gap

Existing literature separately demonstrates multimodal fusion, energy-aware FPGA adaptation, heterogeneous scheduling, uncertainty-aware task allocation and low-precision processing.

The proposed investigation tests the additional value of sensor-derived reliability and tracking uncertainty for runtime assignment of selected perception operators to ARM or FPGA, accounting for data-transfer cost, energy and deadlines.

This remains a candidate contribution. The literature review has not established exclusivity or novelty.

10. Experimental baselines

ID	Strategy	Scheduling inputs
B0	CPU-only	None
B1	Fixed CPU-FPGA	Fixed allocation
B2	Workload-aware CPU-FPGA	Load; latency; communication cost
B3	Uncertainty-aware CPU-FPGA	Load; latency; communication cost; reliability; tracking uncertainty

11. Research hypotheses

ID	Hypothesis	Comparison
H1	Uncertainty-aware scheduling reduces energy relative to workload-aware scheduling without unacceptable tracking-quality degradation	B3 versus B2
H2	Dynamic CPU-FPGA scheduling improves execution efficiency compared with fixed partitioning	B2 versus B1
H3	Reliability-aware allocation improves perception robustness during sensor degradation	B3 versus B2
H4	Scheduling overhead remains small enough to preserve real-time deadline compliance	B3 versus B1 and B2

All hypotheses are untested. No expected percentage savings is represented as a measured result.

12. Evaluation metrics

ID	Metric	Unit
M01	Energy per frame	J/frame
M02	End-to-end latency	ms
M03	Tracking position error	m
M04	Deadline miss rate	%
M05	Scheduling overhead	ms/decision
M06	Hardware transition overhead	ms/transition
M07	Uncertainty calibration	Calibration metric

13. Hardware operation selection

ID	Sensor	Operation	Priority
OP01	LiDAR	ROI filtering	PRIMARY
OP02	LiDAR	Voxel or grid statistics	SECONDARY
OP03	Camera	Grayscale and gradient preprocessing	SECONDARY
OP04	Radar	Range and confidence filtering	OPTIONAL
OP05	Fusion	Object association and tracking	CPU_BASELINE

14. Experimental workflow

Stage	Description	Phase
W01	Dataset preparation	PC
W02	Sensor quality estimation	PC
W03	CPU reference implementation	PC
W04	Scheduler development	PC
W05	FPGA accelerator implementation	HARDWARE
W06	ARM-FPGA communication	HARDWARE
W07	Energy and latency evaluation	HARDWARE
W08	Research analysis	PC_AND_HARDWARE

PC-based stages prepare and validate algorithms. Physical ARM-FPGA experiments are required for hardware energy, transfer overhead and execution results.

15. Scientific limitations

- Many full-text reviews and bibliographic checks remain pending.

- The research gap is a hypothesis, not established novelty.

- No ARM-FPGA performance improvement has yet been demonstrated.

- Sensor and tracking uncertainty require calibration.

- GEMSTAR synchronization and coordinate systems require validation.

- Deadline and tracking-quality thresholds are not yet fixed.

- Physical hardware measurements are required for energy claims.

16. Reproducibility and next steps

The Task_16_Final_Research directory contains the literature CSV, comparison CSV, novelty-risk evidence, hypotheses, metrics, workflow, plotting code, generated figures and this PDF.

Subsequent work must verify remaining publication metadata, read the most relevant full papers, validate sensor data and implement the selected algorithms before drawing experimental conclusions.